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Generalized complex numbers – a common language of diverse geometries

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Abstract. The familiar complex numbers, defined by the relation $i^2 = -1$, provide a powerful bridge between algebra and Euclidean geometry. This article explores a profound generalization: number systems defined by $p^2 = -\theta_0 + p\theta_1$ for real parameters θ_0, θ_1. These generalized complex numbers form a unified framework that naturally describes not only elliptic (Euclidean) geometry but also hyperbolic and parabolic geometries. We explore their algebraic properties and geometric interpretations — showing how "circles" in these systems become conic sections in the Cartesian plane — and their application in defining generalized exponential and trigonometric functions. This synthesis enriches our understanding of the deep connections between algebra and geometry.

Keywords: generalized complex numbers, dual numbers, hyperbolic numbers, affine geometry, generalized Euler formula, Cauchy-Riemann equations.

1 Introduction: Extending the Complex Plane

In mathematics, the need to solve quadratic equations in the 16th century led to the emergence of the concept of imaginary numbers. This idea first appeared in the works of Italian mathematicians Gerolamo Cardano (1501–1576) and his student Rafael Bombelli (1526–1572). By applying the condition that the square of the imaginary unit equals minus one, they attempted to solve algebraic problems.

In the 19th century, Leonhard Euler and other mathematicians began to view complex numbers as an extension of real numbers and explored their full algebraic properties. This led

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to the development of the concept of the complex number plane. A one-to-one correspondence was established between complex numbers and points on this plane. However, within the set of complex numbers, notions such as "greater" and "less" are undefined, since there is no natural ordering among points on the complex plane.

The next step was the emergence of complex-valued functions, whose theory found widespread application not only in mathematics but also in physics, mechanics, and other natural sciences. As a result, new number systems such as double numbers, dual numbers, Hamilton's quaternions, Cayley's octonions, and Clifford and Grassmann numbers were introduced.

Historically, generalized complex numbers have been classified into three groups: ordinary, dual and double numbers, as shown in works [1–3]. These studies examined the geometric connections of dual and hyperbolic numbers.

The theory of generalized complex numbers was revitalized by the work of A. A. Harkin and J. B. Harkin [4], and in recent years, it has been further developed by many mathematicians and physicists. The contributions of A. A. Harkin and J. B. Harkin to the theory of generalized complex numbers are as follows:

- they studied complex units of the form $i^2 = p$ and formulated generalized trigonometry;
- introduced generalized planes (elliptic, hyperbolic, and parabolic) instead of the complex number plane;
- proposed new definitions of angle and modulus;
- derived the generalized Euler formula.

The scope of development for generalized complex numbers can be seen in scientific articles [5–7]. Furthermore, the works [8–10] serve as a logical continuation of the research by A. A. Harkin and J. B. Harkin, focusing on the advancement of the theory of generalized complex-valued functions.

2 Algebraic Foundations and Geometric Motivation

In the two-dimensional Euclidean plane, the position of any point $M(x, y)$ given in the Cartesian coordinate system can be uniquely represented by a complex number $z = x + iy$, $i^2 = -1$. This concept plays a crucial role in aligning geometric and algebraic structures. However, if the plane is equipped with an oblique-angle affine coordinate system, as shown in Figure 1, a relevant question arises: what algebraic expression characterizes the position of the point $M(x, y)$ in such a system?

In Figure 1, the affine coordinate system is defined by the origin O and two linearly independent basis vectors $\vec{e}_1$ and $\vec{e}_2$. These vectors determine the directions of the abscissa and ordinate axes, respectively. Thus, the vector $\overrightarrow{OM}$ is expressed as a linear combination of these basis vectors. Here, x and y are the affine coordinates of the point M.

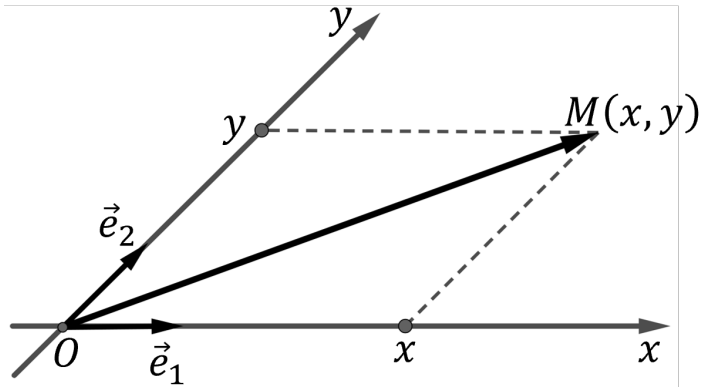


Figure 1. Affine coordinate system on the plane.

The position

$$z = x + py$$

of a point $M(x, y)$ in an affine coordinate system, where

$$p^2 = -\theta_0 + p\theta_1, \qquad \theta_0, \theta_1 \in \mathbb{R}, \tag{1}$$

is uniquely represented by a generalized complex number. Here, the parameter p may be elliptic, parabolic, or hyperbolic in nature. Thus, a one-to-one correspondence is established between the points of the plane and the set of generalized complex numbers.

Special cases

There are three important particular types of generalized complex numbers:

1. $\theta_0 = 1, \theta_1 = 0 \Rightarrow p^2 = -1 \Rightarrow z = x + iy$, $i^2 = -1$ – classical (elliptic) complex numbers;
2. $\theta_0 = 0, \theta_1 = 1 \Rightarrow p^2 = p \Rightarrow z = x + jy$, $j^2 = 1$ – split-complex numbers;
3. $\theta_0 = 0, \theta_1 = 0 \Rightarrow p^2 = 0 \Rightarrow z = x + \varepsilon y$, $\varepsilon^2 = 0$ – dual numbers.

In this extended number system, the following arithmetic operations are defined. Let

$$z_1 = x_1 + py_1, \qquad z_2 = x_2 + py_2, \qquad \lambda \in \mathbb{R}.$$

Then:

$$\begin{aligned} z_1 + z_2 &= (x_1 + x_2) + p(y_1 + y_2), \\ z_1 - z_2 &= (x_1 - x_2) + p(y_1 - y_2), \\ \lambda z &= \lambda x + p\lambda y, \\ z_1 z_2 &= (x_1 x_2 - \theta_0 y_1 y_2) + p(x_1 y_2 + y_1 x_2 + \theta_1 y_1 y_2). \end{aligned}$$

Definition 1. If $x_1 = x_2$ and $y_1 = y_2$, then $z_1 = z_2$.

Definition 2. If a generalized complex number is given as $z = x + py$, where $p^2 = -\theta_0 + p\theta_1$, then the number

$$\bar{z} = x + \theta_1 y - py$$

is called its conjugate, because

$$z\bar{z} = x^2 + \theta_1 xy + \theta_0 y^2 \in \mathbb{R}.$$

Based on this expression, the modulus of a generalized complex number is defined as

$$|z| = \sqrt{x^2 + \theta_1 xy + \theta_0 y^2}. \tag{2}$$

Additionally, the generalized complex numbers z_1/z_2 and z^{-1} (for $z_2 \neq 0$, $z \neq 0$) are defined by

$$\frac{z_1}{z_2} = \frac{x_1x_2 + \theta_0 y_1y_2 + \theta_1 x_1y_2}{x_2^2 + \theta_1 x_2y_2 + \theta_0 y_2^2} + p\,\frac{y_1x_2 - x_1y_2}{x_2^2 + \theta_1 x_2y_2 + \theta_0 y_2^2}, \tag{3}$$

$$z^{-1} = \frac{x + \theta_1 y}{x^2 + \theta_1 xy + \theta_0 y^2} - p\,\frac{y}{x^2 + \theta_1 xy + \theta_0 y^2}. \tag{4}$$

Definition 3. If

$$D = \frac{\theta_1^2}{4} - \theta_0 < 0,$$

then the generalized complex number $z = x + py$, where $p^2 = -\theta_0 + p\theta_1$, is called an *elliptic* number.

The system of elliptic numbers forms a field. In the following sections, we will primarily work within the system of elliptic numbers.

3 The Elliptic Case and Geometric Interpretation

In general, second-order algebraic equations describe various types of curves on the complex number plane. These curves include ellipses, hyperbolas, parabolas, as well as special cases such as straight lines or points.

Accordingly, the set of generalized complex numbers is classified into elliptic, hyperbolic, and parabolic types based on the discriminant of the quadratic form

$$x^2 + \theta_1 xy + \theta_0 y^2 - |z|^2 = 0, \tag{5}$$

where $|z|$ is given by (2).

Definition 4 (Algebraic form)**.** The expression of an elliptic number in the form

$$z = x + py, \qquad p^2 = -\theta_0 + p\theta_1, \quad D < 0, \tag{6}$$

is called its algebraic representation.

To understand the geometric interpretation of elliptic numbers, consider a Cartesian and an affine coordinate system that share a common origin (see Figure 2).

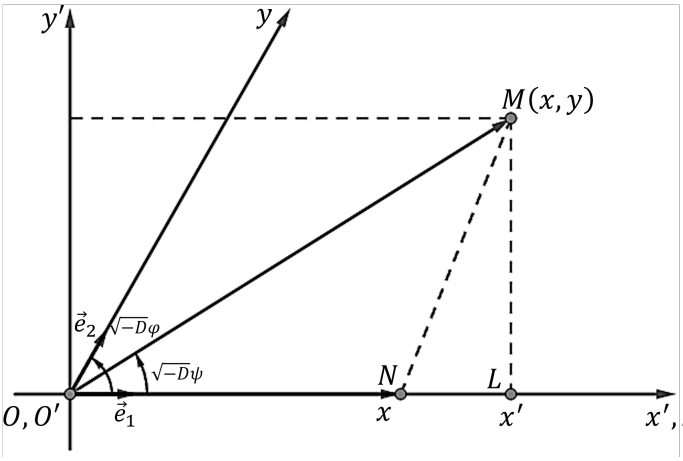


Figure 2. Geometric representation of an elliptic number.

In problem-solving, it is often necessary to transform from the affine coordinate system to the Cartesian coordinate system. According to the definition of an affine transformation, we require that

$$z'_{\text{cart}} = z_{\text{aff}}, \tag{7}$$

where

$$z'_{\text{cart}} = x' + iy', \quad i^2 = -1; \qquad z_{\text{aff}} = x + py, \quad p^2 = -\theta_0 + p\theta_1, \ D < 0.$$

The transformation from the Cartesian coordinate system to the affine coordinate system is defined by the formulas

$$x' = x + \frac{\theta_1}{2}\, y, \qquad y' = \sqrt{-D}\, y. \tag{8}$$

The inverse transformation, i.e. from the affine system to the Cartesian system, is given by

$$x = x' - \frac{\theta_1}{2\sqrt{-D}}\, y', \qquad y = \frac{1}{\sqrt{-D}}\, y'. \tag{9}$$

In these formulas, θ_0 and θ_1 are referred to as the governing parameters of the affine coordinate system.

Definition 5. In the case where $D < 0$, the *elliptic angle* between the real and imaginary axes is defined as the angle $\sqrt{-D}\,\varphi$ between the basis vectors $\vec{e}_1$ and $\vec{e}_2$ of the affine coordinate system.

Remark 6. *The expression $\sqrt{-D}\,\varphi$ represents the total angle, i.e. $\sqrt{-D}\,\varphi = \angle(\vec{e}_1, \vec{e}_2)$.*

Thus, for complex numbers of elliptic type, the angle between their basis vectors is determined by the formula

$$\tan\bigl(\sqrt{-D}\,\varphi\bigr) = \frac{2\sqrt{-D}}{\theta_1}, \tag{10}$$

or

$$\cos\bigl(\sqrt{-D}\,\varphi\bigr) = \frac{\langle \vec{e}_1, \vec{e}_2\rangle}{\|\vec{e}_1\|\,\|\vec{e}_2\|}. \tag{11}$$

In (11), let

$$\vec{e}_1 = (1,0), \qquad \vec{e}_2 = \left(\frac{\theta_1}{2}, \sqrt{-D}\right).$$

Then

$$\langle \vec{e}_1, \vec{e}_2\rangle = \frac{\theta_1}{2}, \qquad \|\vec{e}_1\| = 1, \qquad \|\vec{e}_2\| = \sqrt{\theta_0},$$

and formula (11) becomes

$$\cos\bigl(\sqrt{-D}\,\varphi\bigr) = \frac{\theta_1}{2\sqrt{\theta_0}}. \tag{12}$$

Furthermore, from (12) we derive

$$\sin\bigl(\sqrt{-D}\,\varphi\bigr) = \frac{\sqrt{-D}}{\sqrt{\theta_0}}, \tag{13}$$

since

$$\cos^2\bigl(\sqrt{-D}\,\varphi\bigr) + \sin^2\bigl(\sqrt{-D}\,\varphi\bigr) = 1.$$

The distance between any two points on the plane can be described using elliptic complex numbers. Let the elliptic coordinates of points A and B be denoted by a and b, respectively. Then the distance between them is defined by

$$|AB| = |a - b|.$$

In the plane of elliptic numbers, to understand the geometric meaning of the equation

$$|z - z_0| = r, \tag{14}$$

we use (2), where $z = x + py$, $z_0 = x_0 + py_0$, and $p^2 = -\theta_0 + p\theta_1$ with $D < 0$. Then

$$|z - z_0|^2 = (x - x_0)^2 + \theta_1(x - x_0)(y - y_0) + \theta_0(y - y_0)^2 \equiv r^2. \tag{15}$$

Expanding the brackets and combining like terms with respect to the variables x and y, we obtain a second-order algebraic equation of the form

$$F(x,y) = a_{11}x^2 + 2a_{12}xy + a_{22}y^2 + 2a_1x + 2a_2y + a_0 = 0, \tag{16}$$

where

$$a_{11} = 1, \quad a_{12} = \frac{\theta_1}{2}, \quad a_{22} = \theta_0, \quad a_1 = -(x_0 + \tfrac{\theta_1}{2}y_0), \quad a_2 = -\left(\frac{\theta_1}{2}x_0 + \theta_0 y_0\right), \tag{17}$$

and

$$a_0 = x_0^2 + \theta_1 x_0 y_0 + \theta_0 y_0^2 - r^2. \tag{18}$$

Thus, in the elliptic coordinate system, equations (14) and (16) define a circle centered at the point z_0 with radius r. The question is: what kind of curve does this circle represent in the rectangular Cartesian coordinate system?

To answer this, first expand the equation $|z' - z_0'| = r$ in the Cartesian coordinate system:

$$(x' - x_0')^2 + (y' - y_0')^2 - r^2 = x'^2 + y'^2 - 2x'x_0' - 2y'y_0' + x_0'^2 + y_0'^2 - r^2 = 0. \tag{19}$$

Next, by applying the transformation formulas from the Cartesian system to the affine system,

$$x' = x + \frac{\theta_1}{2}y, \quad y' = \sqrt{-D}\,y, \qquad x_0' = x_0 + \frac{\theta_1}{2}y_0, \quad y_0' = \sqrt{-D}\,y_0, \tag{20}$$

we obtain equation (16). Therefore, whatever curve (16) defines in the Cartesian coordinate system, it corresponds to a circle in the affine system. Thus, we analyze equation (16) under the condition $D = \theta_1^2/4 - \theta_0 < 0$.

We choose the center of symmetry of the ellipse as the origin of the new coordinate system. Let (x_*, y_*) be the center of symmetry of the ellipse defined by (16). The point (x_*, y_*) will be the center of symmetry of the ellipse if and only if its coordinates satisfy the system of equations

$$\begin{cases} \dfrac{\partial F}{\partial x} = a_{11}x_* + a_{12}y_* + a_1 = 0, \\ \dfrac{\partial F}{\partial y} = a_{12}x_* + a_{22}y_* + a_2 = 0. \end{cases} \tag{21}$$

By solving this system using Cramer's rule, we obtain

$$\Delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{vmatrix} = \begin{vmatrix} 1 & \theta_1/2 \\ \theta_1/2 & \theta_0 \end{vmatrix} = \theta_0 - \frac{\theta_1^2}{4} = -D,$$

and

$$\Delta_{x_*} = \begin{vmatrix} -a_1 & a_{12} \\ -a_2 & a_{22} \end{vmatrix}, \qquad \Delta_{y_*} = \begin{vmatrix} a_{11} & -a_1 \\ a_{12} & -a_2 \end{vmatrix}.$$

A straightforward computation shows that

$$\Delta_{x_*} = (-D)\, x_0, \qquad \Delta_{y_*} = (-D)\, y_0,$$

hence

$$x_* = \frac{\Delta_{x_*}}{\Delta} = x_0, \qquad y_* = \frac{\Delta_{y_*}}{\Delta} = y_0. \tag{22}$$

Using (9), these can be written as

$$x_0 = x_0' - \frac{\theta_1}{2\sqrt{-D}}\, y_0', \qquad y_0 = \frac{1}{\sqrt{-D}}\, y_0'.$$

The straight lines that coincide with the principal directions of the ellipse and pass through the point (x_*, y_*) are taken as the coordinate axes of the new coordinate system. In this new coordinate system, the equation of the ellipse can be written in the form

$$\lambda_1 x'^2 + \lambda_2 y'^2 + \frac{I_3}{I_2} = 0, \tag{23}$$

where λ_1, λ_2 are the eigenvalues of the matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} = \begin{pmatrix} 1 & \theta_1/2 \\ \theta_1/2 & \theta_0 \end{pmatrix},$$

and

$$I_1 = a_{11} + a_{22} = 1 + \theta_0, \qquad I_2 = \det A = -D,$$

$$I_3 = \begin{vmatrix} a_{11} & a_{12} & a_1 \\ a_{12} & a_{22} & a_2 \\ a_1 & a_2 & a_0 \end{vmatrix} = Dr^2.$$

Thus, $I_3 = Dr^2$, and its value is negative, since for elliptic numbers $D < 0$. As a result, the conditions $I_2 > 0$, $I_3 < 0$ and $I_1 I_3 < 0$ hold. These conditions show that, in the elliptic coordinate system, the curve defined by

$$|z - z_0| = r, \qquad p^2 = -\theta_0 + p\theta_1, \qquad D = \frac{\theta_1^2}{4} - \theta_0 < 0$$

corresponds to an ellipse in the original Cartesian rectangular coordinate system.

Example 7. *Let us write the equation of the ellipse given in canonical form*

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \tag{24}$$

or

$$x^2 + \frac{a^2}{b^2}y^2 = a^2,$$

with respect to the elliptic coordinate system.

To this end, we compare (24) *with* (16)*. From the comparison, we obtain*

$$a_{11} = 1, \quad a_{12} = 0, \quad a_{22} = \frac{a^2}{b^2}, \quad a_1 = 0, \quad a_2 = 0, \quad a_0 = -a^2.$$

Hence,

$$\theta_1 = 2a_{12} = 0, \qquad \theta_0 = a_{22} = \frac{a^2}{b^2}, \qquad D = \frac{\theta_1^2}{4} - \theta_0 = -\frac{a^2}{b^2},$$

and therefore

$$p^2 = -\theta_0 + p\theta_1 = -\frac{a^2}{b^2}, \qquad p = i\,\frac{a}{b}.$$

Thus, in the elliptic coordinate system determined by the parameters $\theta_0 = a^2/b^2$ and $\theta_1 = 0$, the ellipse (24) *is written as*

$$|x + py| = a, \qquad or \qquad |x + i\,\tfrac{a}{b}y| = a.$$

In addition, equation (24) *can also be obtained as $x'^2 + y'^2 = a^2$ by applying the transformation* (8)*:*

$$x' = x + \frac{\theta_1}{2}y = x, \qquad y' = \sqrt{-D}\,y = \frac{a}{b}y.$$

Example 8. *In an affine coordinate system, consider*

$$5x^2 + 8xy + 5y^2 + 18x + 18y + 9 = 0. \tag{25}$$

Let us study this equation in full. To do so, compare it with (16) *and determine the principal matrix of the second-degree curve*

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix},$$

and the extended matrix

$$\bar{A} = \begin{pmatrix} a_{11} & a_{12} & a_1 \\ a_{12} & a_{22} & a_2 \\ a_1 & a_2 & a_0 \end{pmatrix} = \begin{pmatrix} 5 & 4 & 9 \\ 4 & 5 & 9 \\ 9 & 9 & 9 \end{pmatrix}.$$

1) Compare the invariants:

$$I_1 = 10, \qquad I_2 = \det A = 9, \qquad I_3 = \det \bar{A} = -81, \qquad I_1 I_3 < 0.$$

Hence (25) *represents an ellipse in the Cartesian coordinate system. To write its equation in the elliptic coordinate system, compare* (25) *with* (16). *From this comparison, we obtain*

$$a_{11} = 5, \quad a_{12} = 4, \quad a_{22} = 5, \quad a_1 = 9, \quad a_2 = 9, \quad a_0 = 9.$$

Dividing by 5 *(for normalization) we get coefficients of the form* (17). *Therefore,*

$$\theta_1 = \frac{8}{5}, \qquad \theta_0 = 1, \qquad D = \frac{\theta_1^2}{4} - \theta_0 = -\frac{9}{25} < 0,$$

so

$$p^2 = -\theta_0 + p\theta_1 = -1 + \frac{8}{5}p, \qquad p = \frac{4+3i}{5}, \quad i^2 = -1.$$

2) Next, find the coordinates of the ellipse center via the system

$$\begin{cases} 5x_* + 4y_* + 9 = 0, \\ 4x_* + 5y_* + 9 = 0. \end{cases}$$

Solving gives the center $O'(-1,-1)$. *Thus, in the elliptic coordinate system determined by* $\theta_0 = 1$ *and* $\theta_1 = 8/5$, *the ellipse* (25) *is written as*

$$|(x - x_*) + p(y - y_*)| = r, \qquad |(x+1) + p(y+1)| = \frac{3}{\sqrt{5}},$$

where $p = (4+3i)/5$, $i^2 = -1$.
3) To see which circle's affine image the ellipse (25) *is, use formulas* (19) *and* (20) *to find its center and complex equation. The circle center is*

$$x_0' = x_0 + \frac{\theta_1}{2}y_0 = -1 - \frac{4}{5} = -\frac{9}{5}, \qquad y_0' = \sqrt{-D}\, y_0 = -\frac{3}{5}.$$

Hence, the complex equation of the circle is

$$|z' - z_0'| = r, \qquad |(x' + 9/5) + i(y' + 3/5)| = \frac{3}{\sqrt{5}}.$$

Expanding this equation, the algebraic equation of the circle is

$$x'^2 + y'^2 + \frac{18}{5}x' + \frac{6}{5}y' + \frac{9}{5} = 0.$$

4 The Generalized Exponential and Trigonometric Functions

Many elementary functions can be expressed through Euler's formula. Let us first define the exponential function in the form

$$e^{py} = I(\theta_0, \theta_1, y) + p\,K(\theta_0, \theta_1, y), \tag{26}$$

where $p^2 = -\theta_0 + p\theta_1$.

For (26) the following relations hold:

$$I'(y) = -\theta_0 K(y), \qquad K'(y) = I(y) + \theta_1 K(y). \tag{27}$$

Thus, the functions $I(y)$ and $K(y)$ are solutions of homogeneous linear differential equations of second order with constant coefficients:

$$I''(y) - \theta_1 I'(y) + \theta_0 I(y) = 0, \qquad K''(y) - \theta_1 K'(y) + \theta_0 K(y) = 0.$$

Hence, for generalized complex numbers, the generalized Euler formula takes the form

$$e^{py} = \begin{cases} e^{\frac{\theta_1}{2}y}\left[\left(\cos(\sqrt{-D}\,y) - \dfrac{\theta_1}{2\sqrt{-D}}\sin(\sqrt{-D}\,y)\right) + p\,\dfrac{1}{\sqrt{-D}}\sin(\sqrt{-D}\,y)\right], & D < 0, \\ e^{\frac{\theta_1}{2}y}\left[(1 - \frac{\theta_1}{2}y) + py\right], & D = 0, \\ e^{\frac{\theta_1}{2}y}\left[\left(\cosh(\sqrt{D}\,y) - \dfrac{\theta_1}{2\sqrt{D}}\sinh(\sqrt{D}\,y)\right) + p\,\dfrac{1}{\sqrt{D}}\sinh(\sqrt{D}\,y)\right], & D > 0, \end{cases} \tag{28}$$

where $D = \theta_1^2/4 - \theta_0$.

Now let us take the conjugate of both sides of (26):

$$\overline{e^{py}} = \overline{I(y) + pK(y)},$$

from which it follows that

$$e^{(\theta_1 - p)y} = I(y) + \theta_1 K(y). \tag{29}$$

Multiplying (26) and (29) together yields the fundamental exponential-trigonometric identity for generalized complex numbers:

$$I^2(y) + \theta_1 I(y)K(y) + \theta_0 K^2(y) = e^{\theta_1 y}. \tag{30}$$

As a special case, let $\theta_1 = 0$, $\theta_0 = -1$. Then (30) reduces to the fundamental identity of hyperbolic functions

$$\cosh^2 y - \sinh^2 y = 1,$$

since $D = 1 > 0$, $I(y) = \mathrm{Re}(e^{py}) = \cosh y$, $K(y) = \mathrm{Im}(e^{py}) = \sinh y$.

Now, by means of the generalized Euler formula, we can define the generalized exponential function as

$$e^z = e^{x+py} = e^x\big[I(\theta_0,\theta_1,y) + pK(\theta_0,\theta_1,y)\big].$$

On the elliptic complex plane, the generalized trigonometric functions $\sin z$ and $\cos z$ are expressed via the generalized exponential function as

$$\cos(\sqrt{-D}\,y) = \frac{e^{(\theta_1/2-p)y} + e^{-(\theta_1/2-p)y}}{2}, \qquad \sin(\sqrt{-D}\,y) = \sqrt{-D}\,\frac{e^{(\theta_1/2-p)y} - e^{-(\theta_1/2-p)y}}{2p}. \tag{31}$$

For generalized complex numbers, the following addition formulas hold:

$$\begin{aligned} I(\alpha+\beta) &= I(\alpha)I(\beta) - \theta_0 K(\alpha)K(\beta), \\ K(\alpha+\beta) &= I(\alpha)K(\beta) + I(\beta)K(\alpha) + \theta_1 K(\alpha)K(\beta). \end{aligned} \tag{32}$$

In particular, setting $\alpha = \beta = \varphi$ in (32) yields the double-angle formulas:

$$\begin{aligned} I(2\varphi) &= I^2(\varphi) - \theta_0 K^2(\varphi), \\ K(2\varphi) &= 2I(\varphi)K(\varphi) + \theta_1 K^2(\varphi). \end{aligned} \tag{33}$$

As an application, by means of the generalized Euler formula (28), one can easily find antiderivatives of the integrals

$$\int e^{ax}\sin(bx)\,dx, \quad \int e^{ax}\cos(bx)\,dx, \quad \int e^{ax}\sinh(bx)\,dx, \quad \int e^{ax}\cosh(bx)\,dx,$$

and tabulate them.

To do this, integrate both sides of (26) and use the relations (27). As a result, we obtain formulas for antiderivatives:

$$\begin{aligned} \int I(\theta_0,\theta_1,x)\,dx &= \frac{\theta_1}{\theta_0} I(\theta_0,\theta_1,x) + K(\theta_0,\theta_1,x) + C_1, \\ \int K(\theta_0,\theta_1,x)\,dx &= -\frac{1}{\theta_0} I(x) + C_2, \end{aligned} \tag{34}$$

where C_1, C_2 are constants.

Example 9. *Let* $\theta_1 = 4$, $\theta_0 = 3$. *Then*

$$D = \frac{\theta_1^2}{4} - \theta_0 = 1 > 0.$$

Formula (28) *gives*

$$K(x) = e^{\theta_1 x/2}\,\frac{1}{\sqrt{D}}\sinh(\sqrt{D}\,x).$$

Thus, from (34) *we may write*

$$\int e^{\theta_1 x/2}\sinh(\sqrt{D}\,x)\,dx = -\frac{\sqrt{D}}{\theta_0}e^{\theta_1 x/2}\big(\cosh(\sqrt{D}\,x) - \tfrac{\theta_1}{2\sqrt{D}}\sinh(\sqrt{D}\,x)\big) + C.$$

Hence, in particular,

$$\int e^{2x}\sinh x\,dx = -\frac{1}{3}e^{2x}\big(\cosh x - 2\sinh x\big) + C.$$

5 The Concept of Angle

In the generalized system of complex numbers, the notion of angle is directly related to the problem of geometric description via areas. This approach is similar to defining an angle on the Euclidean plane through the area of a circular sector. However, in generalized cases (such as elliptic or hyperbolic), specific differences arise.

General idea. Let us denote the angle between the ray drawn from the center of the unit ellipse and the real axis by $\sqrt{-D}\,\psi$. Then

$$\sqrt{-D}\,\psi = 2S_{\text{sector}}, \tag{35}$$

where S_{sector} is the area of the sector bounded by the real axis and the ray defining the angle $\sqrt{-D}\,\psi$.

Definition 10. In the elliptic coordinate system, the unit ellipse is defined by the equation

$$|x+py| = 1 \quad \text{or} \quad x^2 + \theta_1 xy + \theta_0 y^2 = 1, \tag{36}$$

where $D = \theta_1^2/4 - \theta_0 < 0$ and $p^2 = -\theta_0 + p\theta_1$.

Remark 11. *Equation* (36) *for the unit ellipse is obtained from the unit circle*

$$x'^2 + y'^2 = 1$$

in the Cartesian coordinate system by applying the transformation (8).

Example 12. *We need to determine the angle and the area corresponding to a point lying on the unit ellipse* (36). *In other words, we must find the angle between the ray passing through a point $z = x + py$ on the unit ellipse and the real axis, as well as the area of the sector between them.*

To find the principal directions of the unit ellipse (36), *we construct the characteristic equation of the matrix*

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix} = \begin{pmatrix} 1 & \theta_1/2 \\ \theta_1/2 & \theta_0 \end{pmatrix}. \tag{37}$$

The characteristic equation is

$$\lambda^2 - \operatorname{Sp} A\,\lambda + \det A = 0 \quad or \quad \lambda^2 - I_1\lambda + I_2 = 0, \tag{38}$$

where

$$I_1 = a_{11} + a_{22} = 1 + \theta_0, \qquad I_2 = \det A = \begin{vmatrix} 1 & \theta_1/2 \\ \theta_1/2 & \theta_0 \end{vmatrix} = -D,$$

and

$$I_3 = \det \bar{A} = \begin{vmatrix} a_{11} & a_{12} & a_1 \\ a_{12} & a_{22} & a_2 \\ a_1 & a_2 & a_0 \end{vmatrix} = \begin{vmatrix} 1 & \theta_1/2 & 0 \\ \theta_1/2 & \theta_0 & 0 \\ 0 & 0 & -1 \end{vmatrix} = D.$$

Remark 13. *For elliptic complex numbers, $D = \theta_1^2/4 - \theta_0 < 0$.*

The eigenvalues of the characteristic equation (38) *are*

$$\lambda_1 = \frac{I_1}{2} + \frac{1}{2}\sqrt{I_1^2 - 4I_2}, \qquad \lambda_2 = \frac{I_1}{2} - \frac{1}{2}\sqrt{I_1^2 - 4I_2}.$$

Thus, the canonical equation of the unit ellipse (36) *can be written as*

$$\frac{x'^2}{1/\lambda_1} + \frac{y'^2}{1/\lambda_2} = 1. \tag{39}$$

From (39)*, the area of the ellipse is*

$$S_{ellipse} = \pi\sqrt{\frac{1}{\lambda_1}\cdot\frac{1}{\lambda_2}} = \pi\frac{1}{\sqrt{\lambda_1\lambda_2}} = \frac{\pi}{\sqrt{I_2}} = \frac{\pi}{\sqrt{-D}}.$$

Hence,

$$S_{sector} = \frac{\pi}{\sqrt{-D}}, \tag{40}$$

and relation (35) *links the angle and the area.*

The direction of the radius vector $\overrightarrow{OM}$ in the Cartesian coordinate system (see Figure 2) is determined by

$$\tan\bigl(\sqrt{-D}\,\psi\bigr) = \frac{\sqrt{-D}\,y}{x + \frac{\theta_1}{2}y}. \tag{41}$$

Formulas (40) *and* (41) *provide the explicit geometric relationship between the elliptic angle and the corresponding area. This connection plays an important role in elliptic trigonometry and in the theory of invariants.*

Example 14. *Let the elliptic coordinate system be defined by the values* $\theta_0 = \theta_1 = 2$. *In this system, the equation*

$$x^2 + 2xy + 2y^2 = 1 \tag{42}$$

represents the unit ellipse.

Next, let us find the angle and the area corresponding to the point $A(0, \sqrt{2}/2)$ *lying on the ellipse* (42). *Since* $\theta_0 = \theta_1 = 2$, *we have*

$$D = \frac{\theta_1^2}{4} - \theta_0 = 1 - 2 = -1, \qquad \sqrt{-D} = 1.$$

Formula (41) *gives*

$$\tan(\sqrt{-D}\,\psi) = \frac{2\sqrt{-D}}{\theta_1} = 1 \quad \Rightarrow \quad \psi = \frac{\pi}{4}.$$

From formula (35), *the corresponding sector area is*

$$S_{sector} = \frac{\sqrt{-D}\,\psi}{2} = \frac{\pi}{8}.$$

More precisely, what we have found here is the area between the real and imaginary axes of the given elliptic coordinate system, since the point $A(0, \sqrt{2}/2)$ *lies on the imaginary axis.*

6 The Importance of the Theory of Generalized Analytic Functions

The theory of generalized complex variable functions is one of the deepest and most widely applied areas of mathematics. At its core lie analytic functions and the Cauchy–Riemann equations that characterize them. The theory of generalized analytic functions not only extends classical complex analysis but also unifies different types of fundamental equations within a common framework.

In this context, in [8], the Cauchy–Riemann equations giving conditions for generalized analytic functions were established in the form

$$\begin{cases} \dfrac{\partial u}{\partial x} + \theta_1 \dfrac{\partial v}{\partial x} = \dfrac{\partial v}{\partial y}, \\ \dfrac{\partial u}{\partial y} + \theta_0 \dfrac{\partial v}{\partial x} = 0, \end{cases} \tag{43}$$

which, in a particular choice of parameters, reduce to the classical Cauchy–Riemann equations given in [4].

Furthermore, the biharmonic operator is expressed as

$$\frac{\partial^4}{\partial z^2\,\partial \bar{z}^2} = \frac{1}{16D^2}\left(\theta_0^2 \frac{\partial^4}{\partial x^4} - 2\theta_0\theta_1 \frac{\partial^4}{\partial x^3 \partial y} + (\theta_1^2 + 2\theta_0)\frac{\partial^4}{\partial x^2 \partial y^2} - 2\theta_1 \frac{\partial^4}{\partial x \partial y^3} + \frac{\partial^4}{\partial y^4}\right). \tag{44}$$

From (44), for $\theta_0 = 1$, $\theta_1 = 0$, we obtain the classical biharmonic operator

$$\frac{\partial^4}{\partial z^2\,\partial \bar{z}^2} = \frac{1}{16}\left(\frac{\partial^4}{\partial x^4} + 2\frac{\partial^4}{\partial x^2 \partial y^2} + \frac{\partial^4}{\partial y^4}\right).$$

Representing the biharmonic operator in the form (44) ensures consistency with classical potential theory. Applications concerning generalized complex numbers and their functions can be found in [9, 10].

7 Conclusion

Generalized complex numbers, defined by $p^2 = -\theta_0 + p\theta_1$, are not merely a mathematical curiosity. They provide a powerful and unified language that:

- *synthesizes geometries*: elliptic, hyperbolic, and parabolic geometries emerge from different choices of parameters θ_0, θ_1;
- *generalizes key concepts*: they support natural extensions of fundamental ideas like the exponential function, trigonometry, and analyticity;
- *has practical utility*: the resulting framework offers elegant solutions to problems in physics and engineering (such as plane elasticity) and provides efficient computational tools for modern algorithms.

This theory demonstrates that the beautiful interplay between algebra and geometry, exemplified by the standard complex numbers, is in fact a single note in a much richer and more harmonious symphony.

Conflict of Interest. The authors declare that they have no conflict of interest.

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Сағындықов Б. Ж., Бимұрат Ж. ЖАЛПЫЛАНҒАН КОМПЛЕКС САНДАР — ӘРТҮРЛІ ГЕОМЕТРИЯЛАРДЫҢ ОРТАҚ ТІЛІ

Классикалық комплекс сандар, яғни $i^2 = -1$ теңдеуімен анықталатын сандар, алгебра мен Евклид геометриясы арасындағы қуатты байланыс көпірі болып табылады. Бұл мақалада нақты параметрлері θ_0 және θ_1 болатын $p^2 = -\theta_0 + p\theta_1$ түріндегі теңдеумен берілетін сан жүйелеріне негізделген терең жалпылау қарастырылады. Мұндай жалпыланған комплекс сандар тек эллиптикалық (Евклидтік) геометрияны ғана емес, сонымен қатар гиперболалық және параболалық геометрияларды да табиғи түрде сипаттайтын біртұтас құрылым түзеді. Мақалада осы сандардың алгебралық қасиеттері мен аффиндік жазықтықтағы геометриялық интерпретациялары талданады — мұнда мұндай жүйелердегі «шеңберлердің» Декарт жазықтығындағы конустық қималарға сәйкес келетіні көрсетіледі. Сонымен қатар, жалпыланған экспоненциал және тригонометриялық функцияларды анықтаудағы қолдану тәсілдері сипатталады. Бұл синтез алгебра мен геометрия арасындағы терең байланыстарды түсінуді байыта түседі.

Түйін сөздер: жалпыланған комплекс сандар, дуал сандар, гиперболалық сандар, аффиндік геометрия, жалпыланған Эйлер формуласы, Коши–Риман теңдеулері.

Сагиндыков Б.Ж., Бимурат Ж. ОБОБЩЁННЫЕ КОМПЛЕКСНЫЕ ЧИСЛА — ОБЩИЙ ЯЗЫК РАЗЛИЧНЫХ ГЕОМЕТРИЙ

Обычные комплексные числа, определяемые соотношением $i^2 = -1$, представляют собой мощный мост между алгеброй и евклидовой геометрией. В данной статье рассматривается глубокое обобщение: системы чисел, задаваемые равенством $p^2 = -\theta_0 + p\theta_1$, где θ_0 и θ_1 — действительные параметры. Эти обобщённые комплексные числа образуют единую математическую структуру, которая естественным образом описывает не только эллиптическую (евклидову), но также гиперболическую и параболическую геометрии. Рассматриваются их алгебраические свойства и геометрические интерпретации — показано, как «окружности» в этих системах превращаются в конические сечения на декартовой плоскости — а также применение при определении обобщённых экспоненциальных и тригонометрических функций. Такой синтез углубляет понимание фундаментальных связей между алгеброй и геометрией.

Ключевые слова: обобщённые комплексные числа, дуальные числа, гиперболические числа, аффинная геометрия, обобщённая формула Эйлера, уравнения Коши–Римана.